Faculty of Engineering and Technology

Electrical and Computer Engineering Department

Artificial Intelligence

ENCS3340

Project 2 Report

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1. Introduction

Image classification is the process of identifying the category of an image based on what it contains. It is a common task in machine learning and is used in many real-world applications like identifying traffic signs, sorting medical images, or recognizing plants and animals.

In this project, we compare three different machine learning models for image classification. These models are Naive Bayes, Decision Tree, and a Feedforward Neural Network using MLPClassifier. Each model works in a different way. Naive Bayes uses probability, Decision Tree follows rule-based decisions, and the neural network learns patterns from the data.

A dataset of images with theme “Nature” was used and divided into three classes: desert, forest, and snow. The goal is to train each model on this data and then test how well it can classify new images. The results were checked using accuracy, precision, recall, F1-score, and confusion matrix.

This project helps us understand how different models perform on the same task and which one gives better results in terms of both accuracy and speed.

2. Dataset Description

The dataset used in this project contains a total of 553 images. Divided onto the three classes.

2.1 Forest Class

There are 182 images in the forest class. These images show different types of forests with trees, plants, and greenery. Some photos are taken during the day while others are taken at night or in low light. Figure 1 below shows what they mostly look like.

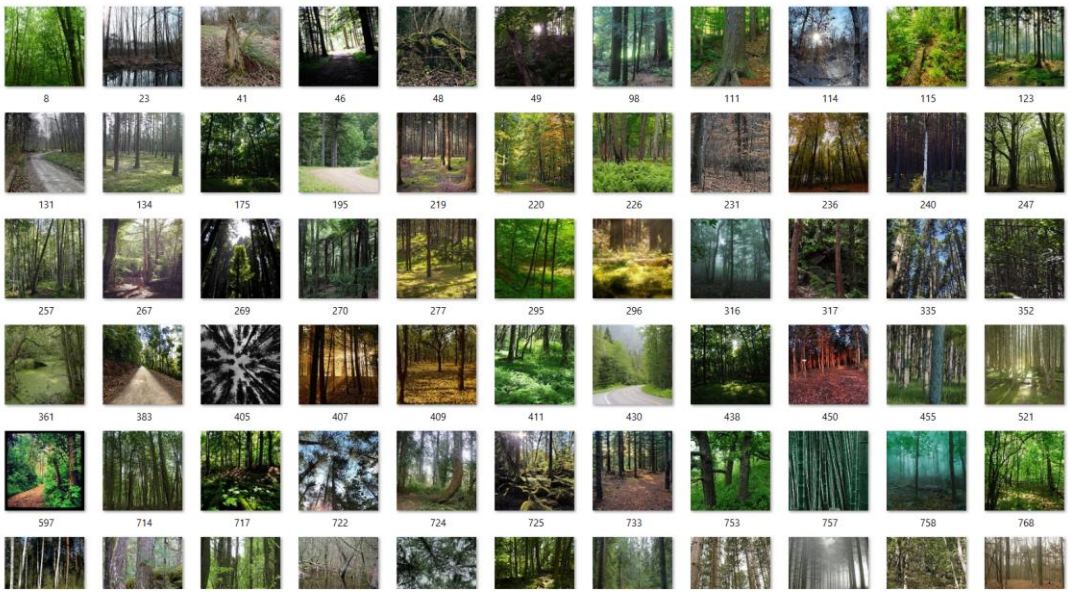

Figure 1: Some forest images

2.2 Desert Class

The desert class has 186 images. These images mostly include sand dunes, dry landscapes, and bright sunlight. Some pictures also show cloudy skies, rocky desert areas, or camels with humans as shown below.

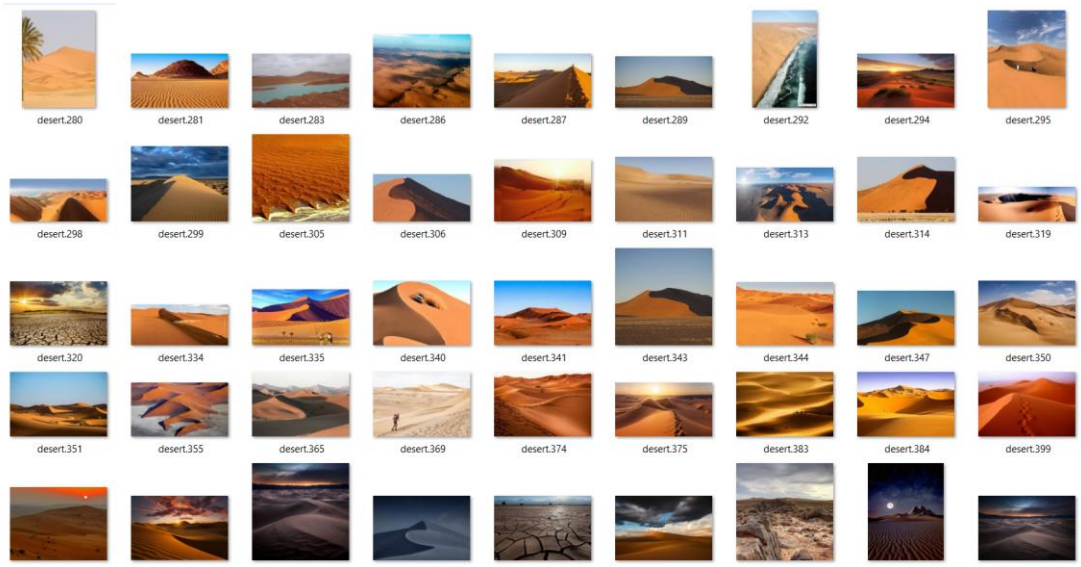

Figure 2: Some desert images

2.3 Snow Class

The snow class contains 185 images. These images show snowy mountains, glaciers, and winter scenes. Some images have people skiing or hiking in snow covered areas.

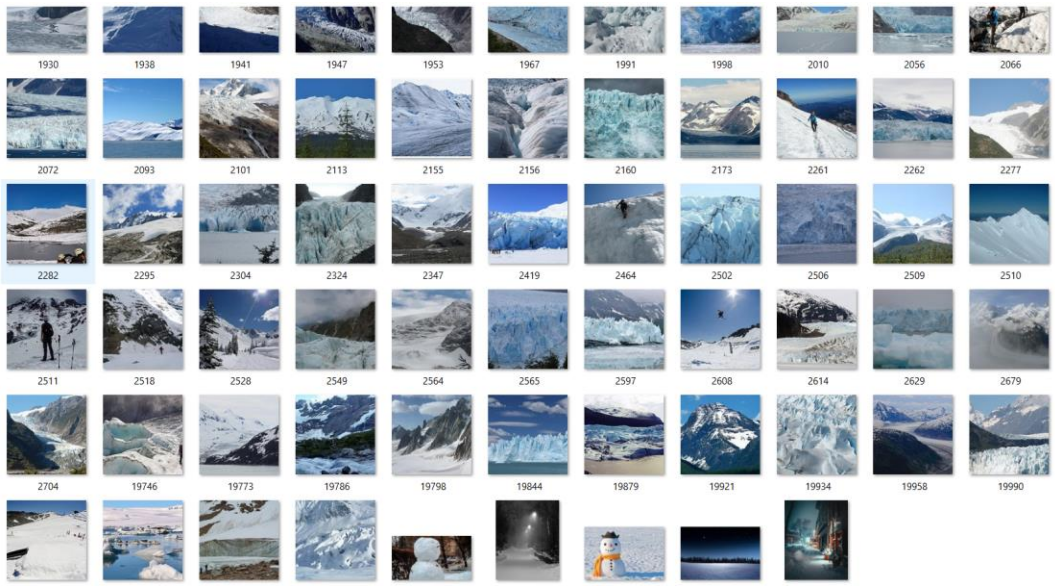

Figure 3: Some snow images

All the images were resized to 32 by 32 pixels and converted to RGB format to make them uniform in size and color format. Then each image was flattened into a one dimensional array and normalized. This step was done to prepare the data so it could be used as input for machine learning models.

The dataset was balanced and diverse. It included both close up and wide angle shots, taken in different lighting conditions and environments. This helped the models learn to identify key features of each class and perform better during classification.

A Google Drive link to the dataset:

https://drive.google.com/drive/folders/1FVUNX-jsDwtXf12ntU5V4Du7lK_hdTlz?usp=sharing

3. Explanation of Each Model

Three different models were used to classify images: Naive Bayes, Decision Tree, and a Feedforward Neural Network. Each model works in a different way and has its own advantages and disadvantages. Below is an explanation of how each one works.

Naive Bayes Classifier

The Naive Bayes model is based on probability. It uses the Bayes theorem and makes a simple assumption that all features (in this case, pixel values) are independent. Even though this assumption is not always true, Naive Bayes still works well in many cases. It is fast to train and works well with high-dimensional data like images. In our code, we used the Gaussian Naive Bayes version, which assumes that the pixel values follow a normal distribution. We used this model as a simple and fast baseline to compare with the other models.

Decision Tree Classifier

The Decision Tree model learns by asking a series of questions about the input features. Each decision splits the data based on the value of one feature and continues until it reaches a final decision, called a leaf. The goal is to build a tree that separates the classes as clearly as possible. The model uses a measure called entropy to decide which features to split on. In our project, we used the entropy-based version and let the tree grow fully. One of the main advantages of decision trees is that they are easy to understand and can be visualized. We also plotted the tree to see how it makes its decisions.

Feedforward Neural Network (MLP Classifier)

The Feedforward Neural Network, also called a Multi-Layer Perceptron (MLP), is a model made up of layers of artificial neurons. Each neuron is connected to all neurons in the next layer. The model learns by adjusting the weights of these connections using a method called backpropagation. In our code, we used two hidden layers with 128 and 64 neurons. The activation function we used is ReLU, and the optimizer is Adam. This model could be more powerful than the other two and can learn more complex patterns in the data, but it needs more time and data to train properly.

4. Evaluation Results

To compare the performance of the three models, we measured their accuracy as well as the precision, recall, and F1-score for each class. The dataset was divided into 75% train set and 25% test set. The test set included images from the three categories: desert, forest, and snow. We also used confusion matrices to understand which classes were being confused with others.

Overall Accuracies:

- Naive Bayes: 0.7986
- Decision Tree: 0.8201
- MLP Neural Network: 0.8633

These results show that the neural network (with 2 hidden layers) performed the best overall, followed by the decision tree and then the Naive Bayes classifier.

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Accuracies:

Naive Bayes     : 0.7986
Decision Tree   : 0.8201
MLP Neural Net  : 0.8633

==================================
Naive Bayes Results
==================================
Accuracy: 0.7986

Class: Desert
Precision: 0.76
Recall   : 0.68
F1-score : 0.72

Class: forest
Precision: 0.84
Recall   : 0.83
F1-score : 0.84

Class: snow
Precision: 0.79
Recall   : 0.89
F1-score : 0.84
```

```
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==================================
Decision Tree Results
==================================
Accuracy: 0.8201

Class: Desert
Precision: 0.90
Recall   : 0.79
F1-score : 0.84

Class: forest
Precision: 0.76
Recall   : 0.85
F1-score : 0.80

Class: snow
Precision: 0.81
Recall   : 0.83
F1-score : 0.82
```

Figure 4: Evaluation results (1)

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MLP Neural Net Results
==================================
Accuracy: 0.8633

Class: Desert
Precision: 0.89
Recall   : 0.89
F1-score : 0.89

Class: forest
Precision: 0.83
Recall   : 0.76
F1-score : 0.80

Class: snow
Precision: 0.86
Recall   : 0.93
F1-score : 0.90
```

Figure 5: Evaluation results

The precision, recall, and F1-score values show how well each model performed for each class. For the desert class, the neural network gave the best results with high precision and recall, while the decision tree also performed well. Naive Bayes had lower recall, meaning it missed more desert images. For the forest class, all models gave similar F1-scores, but Naive Bayes had slightly better balance between precision and recall. For the snow class, the neural network again gave the best performance, with high recall and precision, showing it was very accurate in detecting snow images.

Confusion Matrices

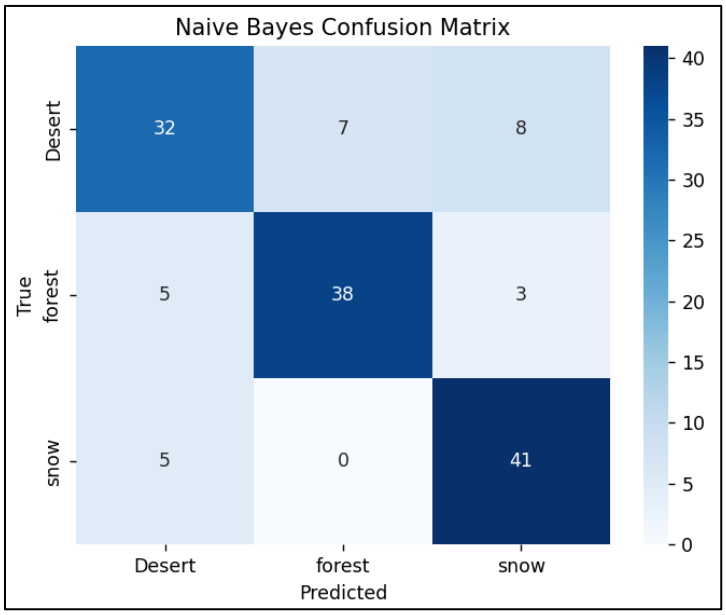

Figure 6: Naive Bayes confusion matrix

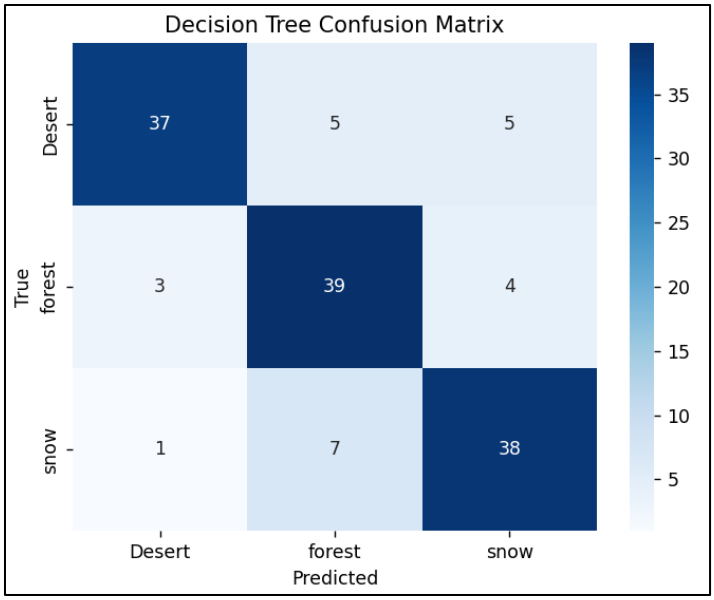

Figure 7: Decision tree confusion matrix

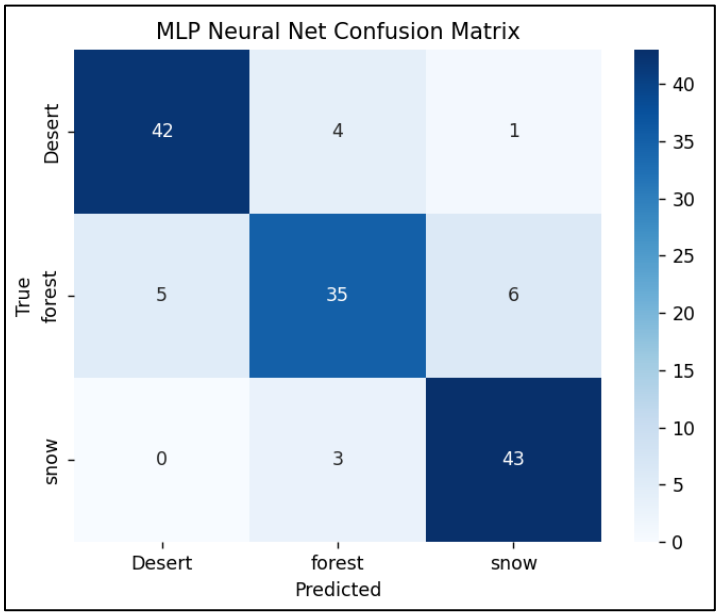

Figure 8: MLP confusion matrix

The confusion matrices provide a visual comparison of how each model performed on the test set. The MLP neural network with two hidden layers had the strongest results, with the highest number of correct predictions for all classes and very few misclassifications. It perfectly classified all snow images except for three that were labeled as forest. The decision tree also performed well, especially for forest and snow, but had slightly more confusion between the classes, especially snow misclassified as forest. The Naive Bayes model showed more errors overall, particularly with desert images, where several were confused with forest and snow. It also misclassified a few snow images as desert.

We also tried having different numbers of hidden layers with different neurons. For example, we had 1 layer with 128 neurons (instead of 2 layers with 128 and 64 neurons) in the MLP Neural Network model and got the following results:

```
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MLP Neural Net Results
==================================
Accuracy: 0.8201

Class: Desert
Precision: 0.79
Recall   : 0.87
F1-score : 0.83

Class: forest
Precision: 0.88
Recall   : 0.65
F1-score : 0.75

Class: snow
Precision: 0.81
Recall   : 0.93
F1-score : 0.87
```
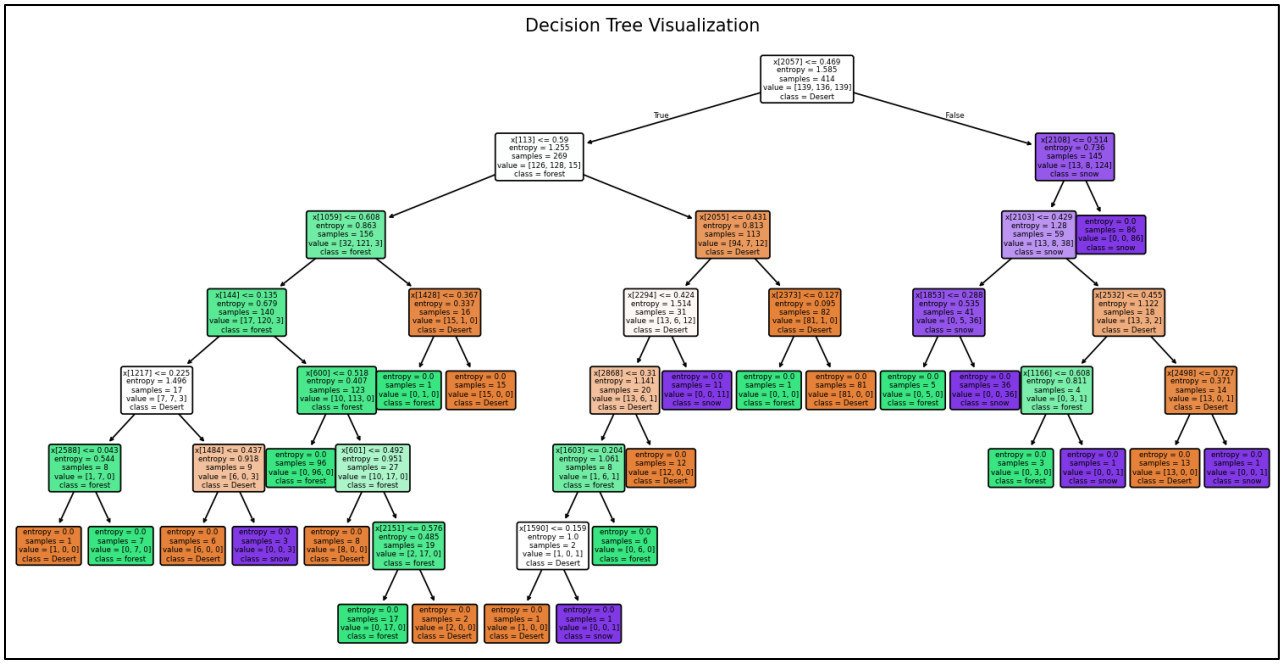

Figure 9: MLP results with 1 hidden layer

With this change, the MLP accuracy dropped from 0.8633 to 0.8201. This new accuracy is the same as the decision tree model. The desert and snow classes still performed well, especially snow with a recall of 0.93. However, the forest class showed a drop in recall to 0.65, even though its precision increased to 0.88. This means the model became more selective in predicting forest but missed many actual forest images.

We also found out that having 1 layer with 64 neurons gave the same accuracy as having2 layers with 128 and 64 neurons but with different precision, recall, and f1-score. This shows that accuracy alone does not always tell everything. A model with fewer neurons may still perform well overall but can behave differently on each class. The reason for this is that network depth and size affect how the model learns patterns. Smaller networks may generalize differently or become more confident in fewer predictions, while deeper networks can capture more detailed patterns but may overfit or require more data to perform consistently across all classes.

Decision Tree Visualization



The decision tree visualization shows how the model makes its classification decisions by splitting the data based on pixel values. Each node represents a condition that divides the dataset, aiming to reduce uncertainty using entropy. The tree branches out until it reaches leaves that represent final class predictions. The colors show the predicted class at each node, with green for forest, orange for desert, and purple for snow. The structure of the tree reveals that the model was able to form distinct rules for classifying many samples, but also includes some complex and deep branches, which may indicate overfitting to specific patterns in the training data.

5. Comparative Analysis and Discussion

The MLP neural network with two hidden layers performed the best overall because it was able to learn complex patterns and relationships in the image data. Its deeper structure allowed it to separate the classes more clearly, especially for snow and desert, resulting in fewer misclassifications. When the MLP was reduced to a single hidden layer, its accuracy dropped to match that of the decision tree, showing that fewer layers limited its learning ability. However, this is not always true for all MLPs, as adding more layers can sometimes lead to overfitting. The decision tree gave good results and was able to handle the task well by using simple rule based splits. However, it made more mistakes between visually similar classes, like forest and snow, because it cannot model complex feature interactions. Naive Bayes had the lowest performance because it relies on the assumption that all pixel features are independent, which is not true for image data. This made it less effective, especially in identifying desert images, and led to more overall errors. These results highlight the strengths of neural networks for image tasks and the trade offs between accuracy and model simplicity.